

CAPSTONE PROJECT

Mutual Fund Analytics Platform

End-to-End Data Engineering, ETL Pipeline & Interactive Dashboard

Transforming raw mutual fund data into actionable investment intelligence.

Sai Khedekar

The mutual fund ecosystem generates large volumes of fragmented data, making performance evaluation and investor analysis difficult.

Challenges

- Disconnected datasets
- Limited investor analytics
- Lack of integrated performance evaluation
- No unified dashboard for decision making

Objectives

- Build end-to-end ETL pipeline
- Clean and standardize data
- Perform EDA
- Calculate performance metrics
- Develop interactive dashboards
- Generate business insights

 From raw data to investment intelligence.

10 integrated datasets powering the analytics platform

Fund Master	NAV History
AUM by Fund House	Monthly SIP Inflows
Category Inflows	Industry Folio Count
Scheme Performance	Investor Transactions
Portfolio Holdings	Benchmark Indices

2022–2026

Historical Period

40+

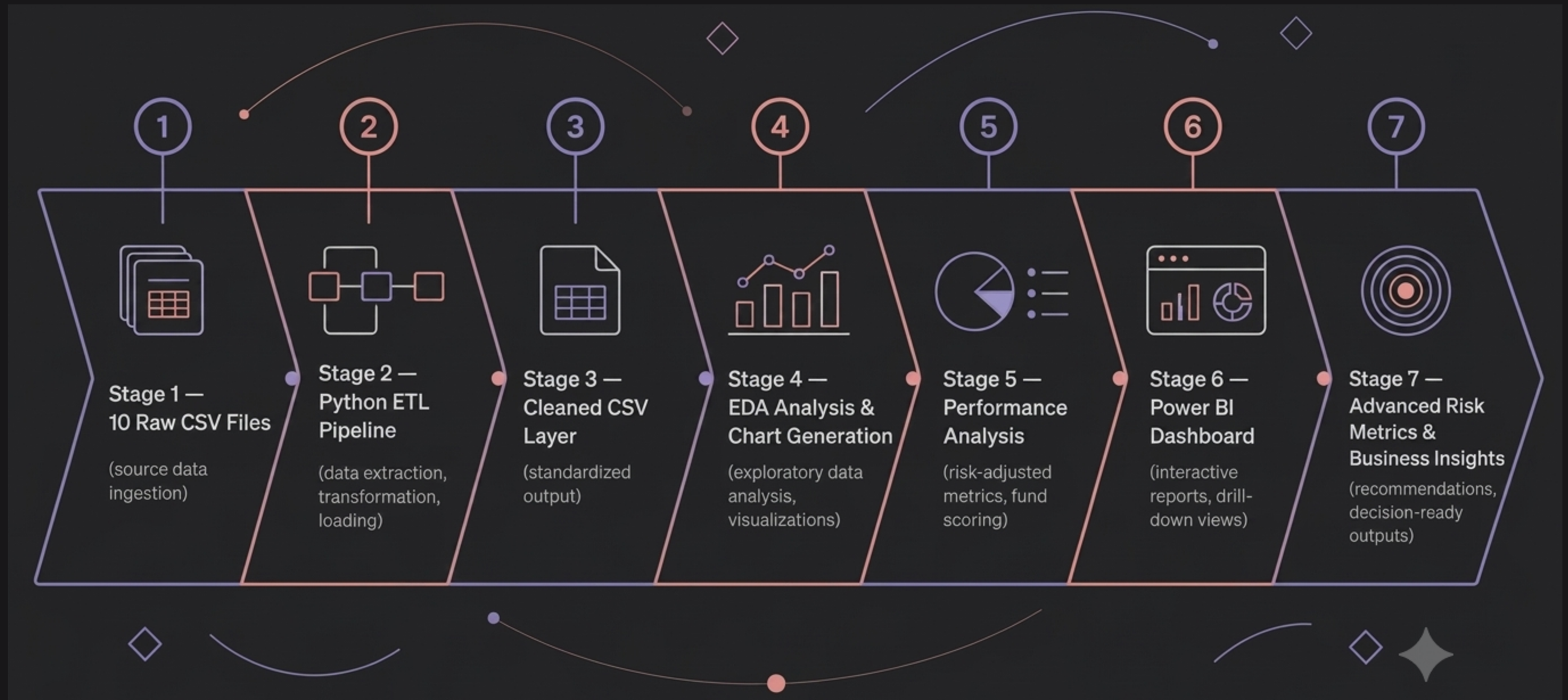
Schemes Analyzed

10

Integrated Datasets

★ The analysis combines historical mutual fund datasets with live NAV data retrieved through the MFAPI API to ensure up-to-date market insights.

End-to-End Analytics Architecture



❏ A modular analytics pipeline transformed raw data into decision-ready insights.

Key Statistics: Industry Snapshot

41%

Young Investors

Investors belong to the 26–35 age group

66%

Urban Activity

Investment activity originates from T30 cities

SBI

Top AMC

Leads the industry by AUM

₹31K+ Cr

Monthly SIP Inflows

Increased from ₹11,500 Cr to ₹31,000+ Cr

9+ Cr

Active SIP Accounts

Grew from 5 Cr to 9+ Cr

50%+

SIP Growth

Exceeded 50% YoY during 2024

↑ 74%

Industry NAV

Increased by approximately 74%

32%

Portfolio Allocation

Banking and IT account for ~32%

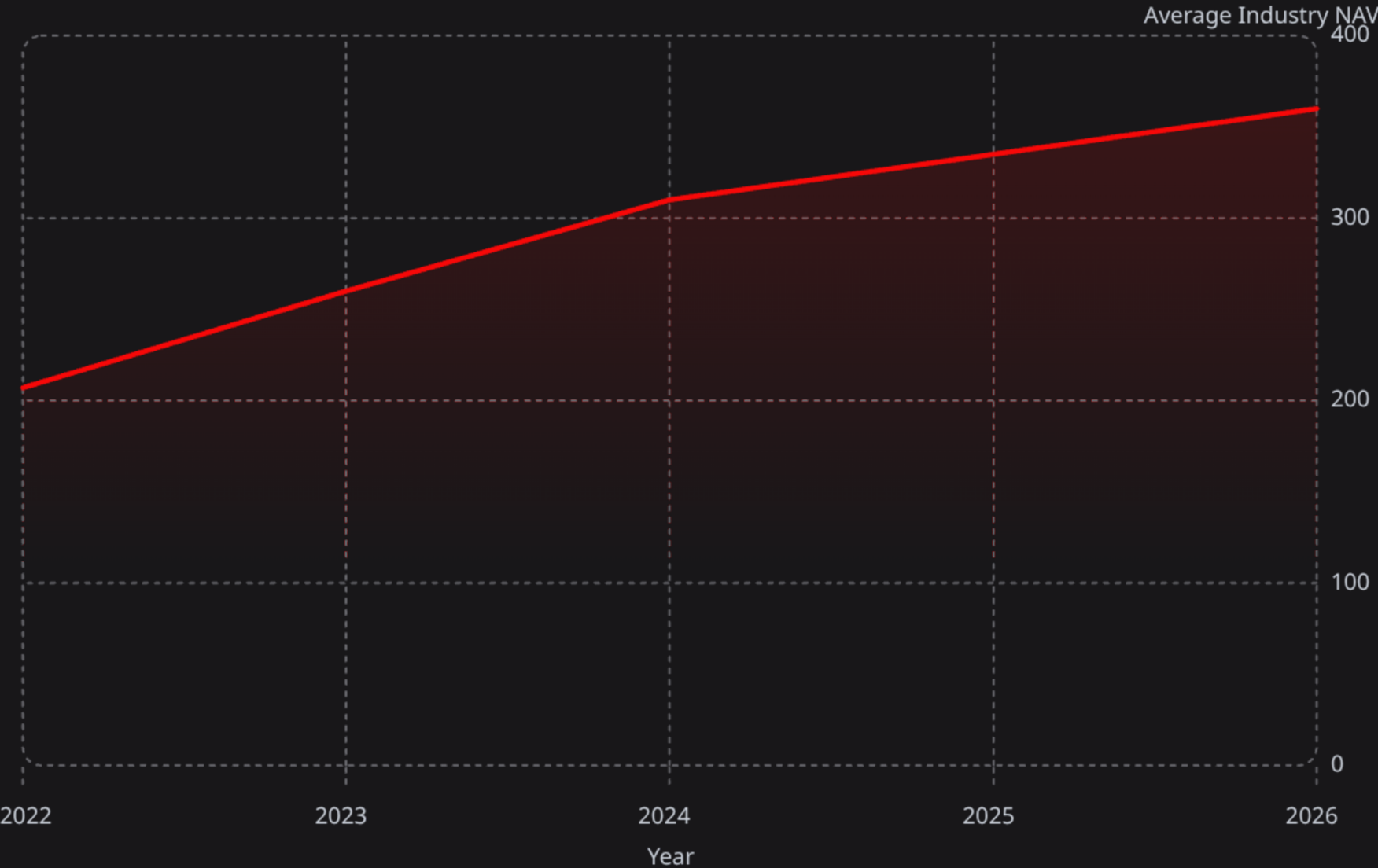
Low

Fund Correlations

Indicate strong diversification opportunities

Industry Growth Remained Strong Throughout the Study Period

Average Industry NAV
Trend



Average Industry NAV increased from 207 in 2022 to nearly 360 by 2026, reflecting strong long-term appreciation across the mutual fund industry.

Key Numbers

Monthly SIP Inflows
₹11,500 Cr → ₹31,000+ Cr
📈 +170% Growth



Industry NAV
207 → 360
📈 +74% Growth

Active SIP Accounts
5 Cr → 9+ Cr
📈 +80% Growth



Retail participation continues to accelerate across the mutual fund ecosystem. Rising SIP inflows (+170%), expanding active SIP accounts (+80%), and sustained NAV appreciation (+74%) indicate growing investor confidence and long-term industry expansion.

The mutual fund industry experienced strong growth during the study period. Average NAV increased by 74%, reflecting long-term value creation. Monthly SIP inflows grew by approximately 170%, while active SIP accounts expanded by 80%, demonstrating increasing retail investor participation. Together, these indicators show a healthy and rapidly expanding mutual fund ecosystem.

Risk-Adjusted Performance Framework

Traditional returns do not capture the full investment picture. Risk-adjusted metrics were used to evaluate fund quality and consistency.



CAGR

Measures long-term growth



Sortino Ratio

Measures return relative to downside risk



Beta

Measures market sensitivity



Sharpe Ratio

Measures return generated per unit of risk



Alpha

Measures benchmark outperformance



Maximum Drawdown

Measures worst historical decline

Framework Applied Across 40+ Mutual Fund Schemes

Metric	Purpose
CAGR	Growth Measurement
Sharpe	Risk Efficiency
Sortino	Downside Protection
Alpha	Excess Return
Beta	Market Exposure
Drawdown	Capital Preservation

Key Evaluation Principle

- ✓ Higher CAGR
- ✓ Higher Sharpe Ratio
- ✓ Higher Sortino Ratio
- ✓ Positive Alpha
- ✓ Beta close to 1
- ✓ Lower Drawdown



Key Business Insight:

Risk-adjusted evaluation identified funds that delivered superior returns without taking excessive risk. This approach provides a more reliable framework for fund selection than relying on returns alone.

Top Funds Generated Strong Risk-Adjusted Returns

Funds were ranked using CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, and Maximum Drawdown.

 Top Performing Fund

Mirae Asset Large Cap Fund

Metric	Value
3-Year CAGR	≈ 34%
Sharpe Ratio	≈ 1.45
Alpha	≈ 0.27
Risk Profile	Moderate
Ranking	#1

Rank	Fund
1	Mirae Asset Large Cap Fund
2	ICICI Prudential Bluechip Fund
3	HDFC Mid Cap Opportunities Fund
4	Kotak Flexicap Fund
5	ICICI Prudential Midcap Fund

40+ Funds Analyzed

6 Risk Metrics Evaluated

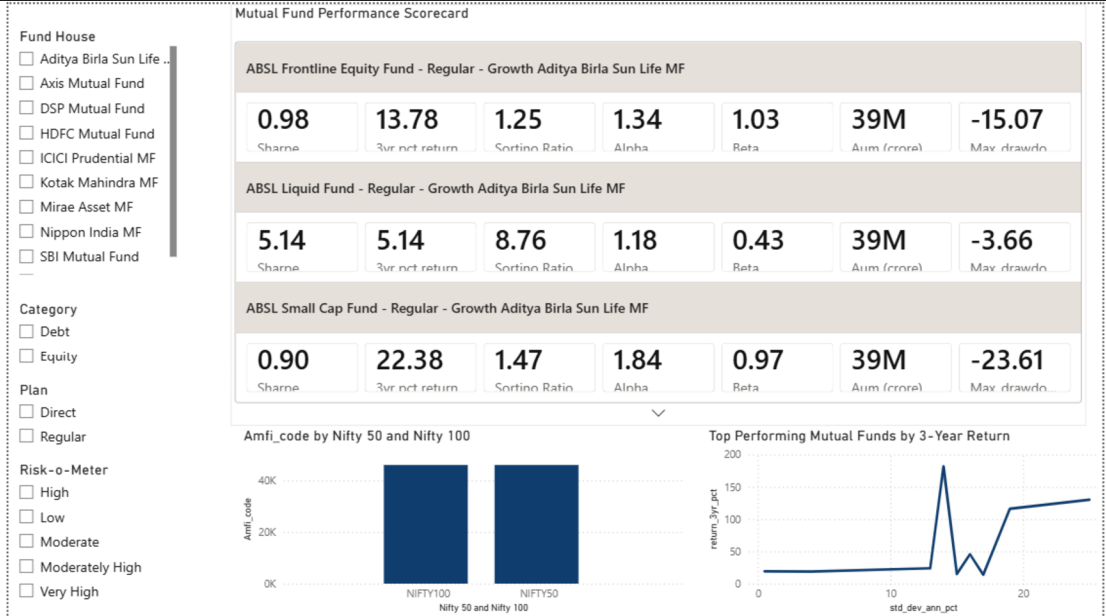
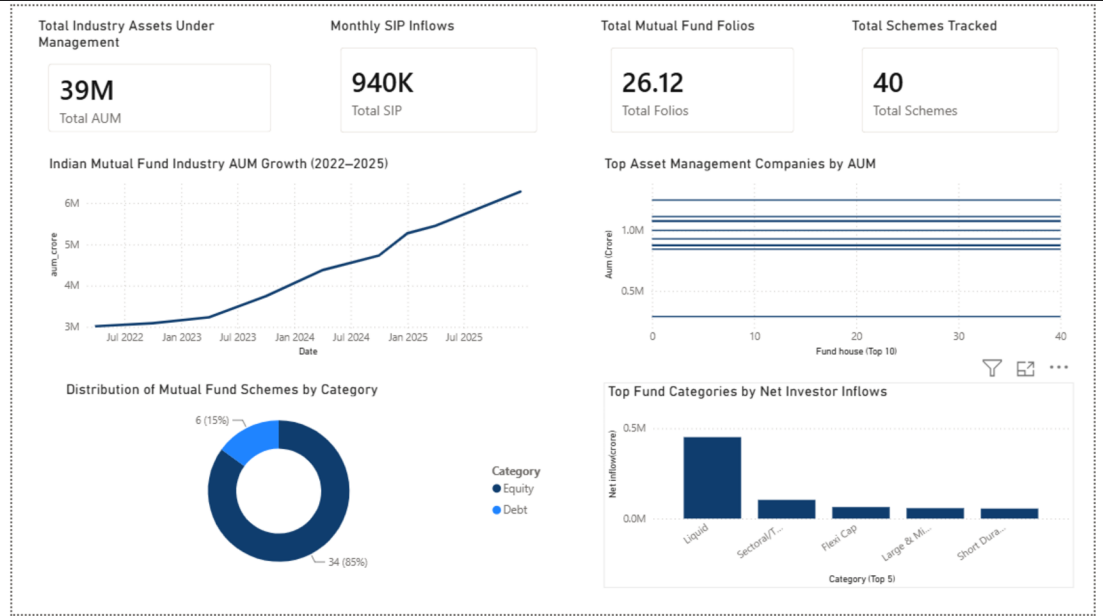
2022–2026 Study Period

 **Key Business Insight:**

Mirae Asset Large Cap Fund emerged as the highest-ranked scheme, delivering approximately 34% CAGR while maintaining strong risk-adjusted performance. Multiple actively managed funds outperformed benchmark indices over the three-year evaluation period.

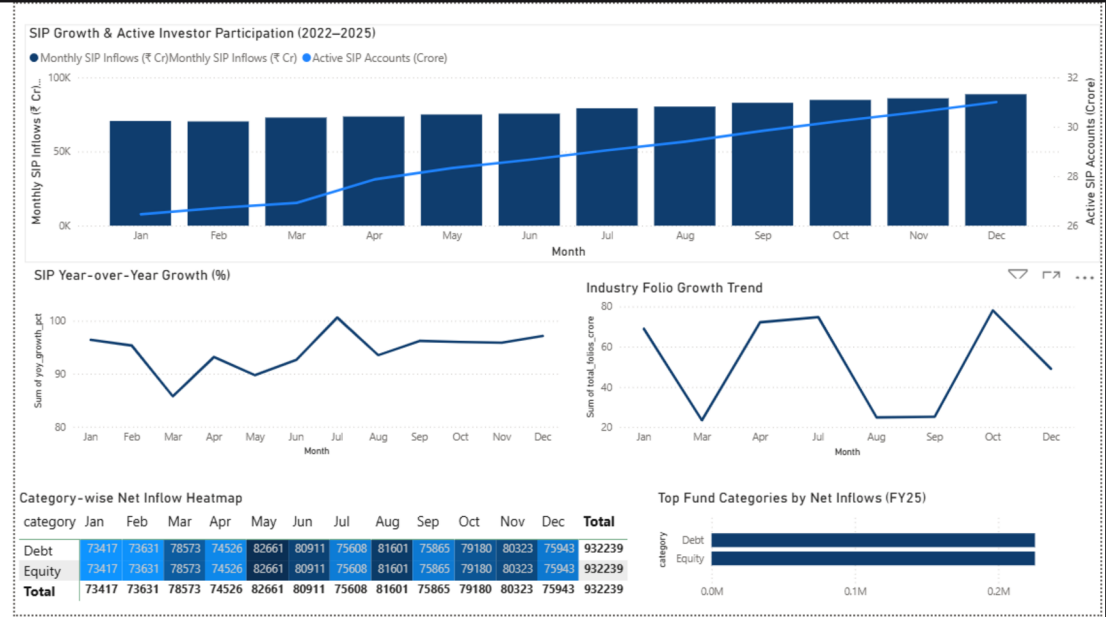
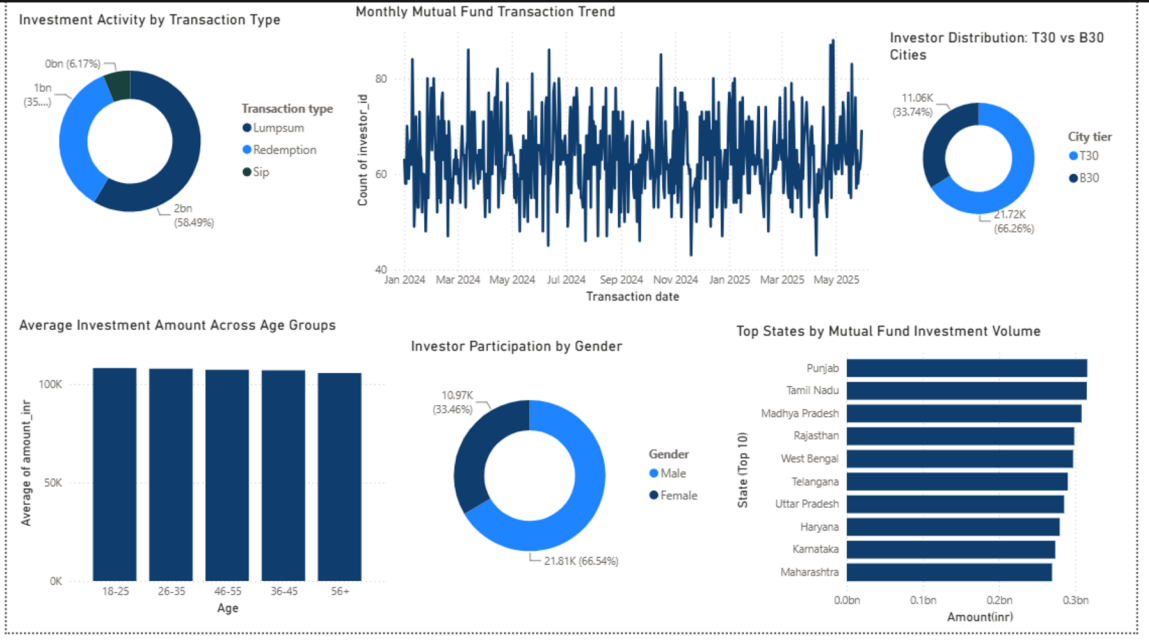
Risk-adjusted analysis revealed that top-performing funds consistently generated higher returns while maintaining controlled volatility, proving that superior performance is driven by both growth and risk management rather than returns alone.

Power Bi Dashboard Snapshots



Key KPIs Tracked Dashboard Capabilities

- Industry-wide growth tracking
- AMC comparison
- Category analysis
- Market trends



Provides a macro view of industry performance and growth.

Total AUM 39M

Total SIP Inflows 940 k

Total Folios 26.12

Total Schemes 40

Interactive Decision-Making Dashboards

Power BI dashboards enable stakeholders to analyze industry trends, fund performance, investor behavior, and market growth through a unified i

Dashboard Capabilities



Dashboard Coverage Table

Pages	Coverage
Page 1	Industry Overview
Page 2	Fund Performance
Page 3	Investor Analytics
Page 4	SIP & Market Trends

i Key Business Insight:
Business users can seamlessly transition from industry-level trends to individual fund analysis, enabling faster and more informed investment decisions through interactive exploration.

Key Insights from the Analysis



Key Findings

- Young professionals dominate investing
- T30 cities drive participation
- SIP adoption is accelerating
- SBI Mutual Fund leads industry AUM
- Banking & IT dominate portfolio allocations
- Liquid funds attract highest inflows
- Top AMCs control a large share of assets

Recommendations

- Expand B30 penetration
- Increase female investor participation
- Strengthen SIP-focused campaigns
- Promote diversification education
- Develop products targeting younger investors

❏ **India's mutual fund industry continues to expand through growing retail participation and systematic investing.**

Thank You

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https://github.com/saiKhedekar/CapstoneProject_1

Building data-driven solutions for smarter investment decisions.